Lamina

[ABSTRACT]

Technical Validation Report: ML-Enhanced Solar Performance Analysis

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Prepared by: George Cooper, Founder, Lamina

*This document serves as an **Executive Abstract** for the **43-page Technical Validation Report**. It contains the Summary, Commercial Findings, and the original Table of Contents to demonstrate the depth of the underlying analysis. The full 43-page validation document is available upon request.*

1.0. Executive Summary

1.1. Objective:

This report details the technical validation of Lamina's proprietary hybrid photovoltaic (PV) modelling framework. The objective of the analysis was to assess the model's accuracy in predicting energy yield and its capability to act as a "Digital Twin" for operational assets. By combining on-site measurements with satellite irradiance data and physics-enhanced machine learning (ML), the model establishes a calibrated performance baseline. This approach aims to transition asset management from reactive monitoring to proactive quantification of specific loss mechanisms.

1.2. Model Description:

The analysis utilises a dual-layer modelling approach designed to minimise uncertainty in yield predictions:

- **Physics-Based Layer:** A deterministic model generates a theoretical benchmark (Y_{phys}) based on site-specific hardware parameters, system geometry and real-time meteorological inputs.

- **Machine Learning (ML) Layer:** A secondary ML algorithm, trained on historical SCADA and weather data, corrects, and supplements the physics baseline. This layer captures non-linear, site-specific variables such as horizon shading, complex soiling accumulation rates, and thermal dissipation variances.

1.3. Key Findings:

The model underwent blind 5-fold cross-validation against operational data from the Newquay case study asset (Analysis Period: 2,742 days). To rigorously assess diagnostic capability, the model was tested against two distinct data sub-sets: a "Golden" dataset (representing healthy operation) and the "Full" dataset (including all operational anomalies).

Table 1: Hybrid ML Model Performance Metrics

Metric	Validation Scenario	Value	Interpretation
Digital Twin Precision (R^2)	Golden Dataset	0.987	Under healthy conditions, the model captures >98% of performance variance, confirming high-fidelity calibration.
Operational Reality (R^2)	Full Dataset	0.936	The divergence from the Golden baseline (0.98 vs. 0.93) successfully isolates non-linear fault signals.
nRMSE	Full Dataset	5.68%	Mean error magnitude relative to site capacity (inclusive of anomalies)
Performance Gain vs. Persistence			
Physics-Enhanced ML		+82.8%	Significant reduction in forecast error compared to naïve baselines.
ML Value-Add vs. Physics Only		+41.8%	The ML layer resolved 41.8% of the residual error left by the physics model alone.

1.4. Business Outcomes

The application of the Physics-Enhanced ML model on the case study asset facilitated a granular decomposition of system losses (Figure 1). The analysis validates three critical diagnostic capabilities.

- **Systematic Loss Quantification:** Contrasting generic Performance Ratio (PR) calculations, the generated Energy Waterfall explicitly quantifies specific yield inhibitors. Major losses identified include Degradation (4.3%), Inverter inefficiencies (3.2%), and soiling (1.0%), allowing for precise CapEx/OpEx cost-benefit analysis.

- **Signal-to-Noise Filtration:** The methodology employs statistical significance testing and additional correlation analysis to separate physical loss signals from sensor noise. Reported losses are validated within a 95% Confidence Interval ($p < 0.05$), ensuring that maintenance resources are not deployed for stochastic modelling artifacts.

- **Sub-System Fault Isolation:** Beyond system-level averages, the analysis demonstrated the capability to isolate granular hardware performance. The model identified statistically divergent behaviour in String 2 compared to the array median. This granular visibility enables the transition from broad, untargeted string testing to specific, evidence-based warranty monitoring of individual sub-components.

Conclusion: The analysis identifies a confirmed recoverable yield potential of 2-3% annually. Additionally, successful isolation of string-specific performance divergence demonstrates the critical ability to translate complex data into quantifiable revenue opportunities, supporting a strategic transition to Condition-Based Maintenance (CBM).

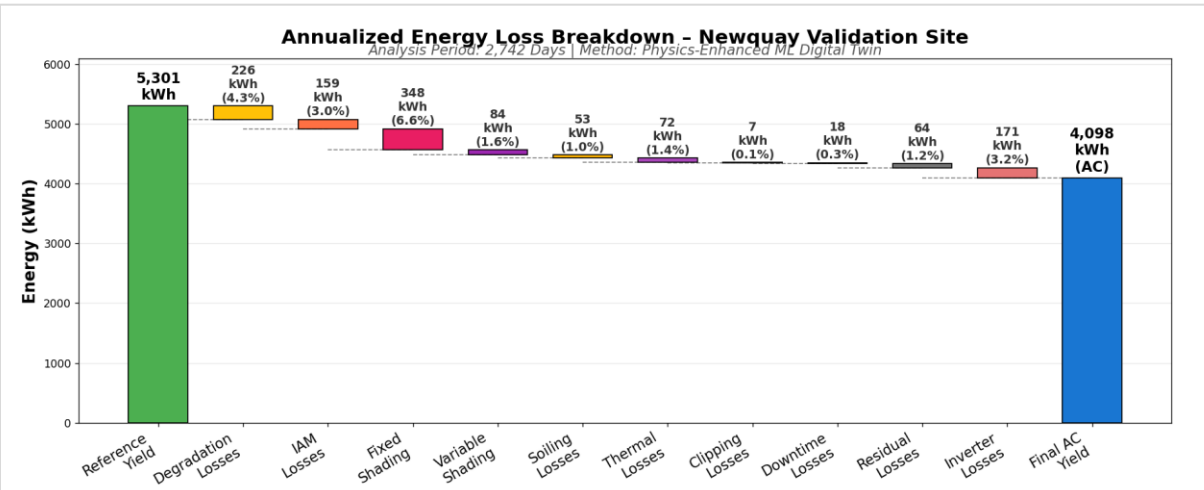

Figure 1: Energy Loss Waterfall

Figure 1: Yearly Average Energy Loss Waterfall. Analysis Period: 2,742 Days. The chart details the transition from Reference Yield (Y_{ref}) to Final AC Yield (Y_f). Average residual loss (1.2%)

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Full 43-page technical report available upon request.